

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI
COLLEGE OF ENGINEERING

**BSC CIVIL ENGINEERING, BSC INDUSTRIAL ENGINEERING,
BSC TELECOMMUNICATIONS ENGINEERING**

SECOND SEMESTER PRACTICE EXAMINATION (2021/2022)

FIRST YEAR (UNDERGRAD)

MATH 152—CALCULUS WITH ANALYSIS

SEPTEMBER 2022

TIME ALLOWED: 2 HRS

Your index number:

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Programme of Study: _____

These are practice questions; the main exams will be similar.

Read the following instructions carefully;

1. Write your index number and programme of study **CLEARLY** at the top of this page in the spaces provided.
2. Answer **ALL** Questions by shading the correct option on the **SCANNABLE SHEET** and **CIRCLE** your answers with pen in the question paper.
3. If you believe an answer you obtain to a question is not in the multiple-choice answers, kindly **SHADE** the letter (E) on the scannable sheet and **WRITE** your answer against the question on the question number paper.
4. No communication without explicit permission from invigilation team.
5. Do not take any examination materials from the Examination Hall.

Part I: Limits and Continuity

Evaluate the following limits.

(1) $\lim_{x \rightarrow 0} \tan^{-1} \frac{1}{x}$

- A. 0 B. 8 C. -11 D. Does not exist.

(2) $\lim_{x \rightarrow \infty} \frac{(2x-3)(3x+5)(4x-5)}{3x^3+x-1}$

- A. 8 B. $\frac{1}{\pi}$ C. -4 D. e

(3) $\lim_{x \rightarrow 0} \frac{\sqrt{x+3}-\sqrt{3}}{x}$

- A. $\sqrt{3}$ B. 0 C. $\frac{1}{2\sqrt{3}}$ D. The limit does not exist

(4) $\lim_{x \rightarrow \infty} \frac{x+2}{9x^2+1}$

- A. 0 B. $\frac{1}{9}$ C. $\frac{2}{9}$ D. ∞

- (5) What value of $f(0)$ should be chosen so that the function below is continuous at $x = 0$

$$f(x) = 1 - x \sin \frac{1}{x}$$

- A. 1 B. 2 C. 4 D. 8

- (6) What expression is the limit below?

$$\lim_{h \rightarrow 0} \frac{f(x+2h) - 2f(x+h) + f(x)}{h^2}$$

- A. $f(x)$
B. $f'(x)$
C. $f''(x)$
D. None of the above

Use the following to answer the next **THREE** questions.

- A. $e(x) = |x|$ B. $f(x) = 0$ C. $g(x) = \frac{x}{|x|}$ D. $h(x) = x + |x|$

- (7) Which of the functions is differentiable?
- (8) Which of the functions is not defined for all real numbers?
- (9) Which of the functions can never be zero as x approaches 0?
- (10) Which of the statements BELOW is neither true nor false?
- A. If $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists, then f is continuous at $x = a$.
- B. If f is continuous at $x = a$, then it is differentiable there.
- C. If $f'(a)$ exists, then $\lim_{x \rightarrow a} f(x)$ exists as well.
- D. If $\lim_{x \rightarrow a} f(x)$ exists, $f'(a)$ may not exist.

Part II—Differentiation and Differentiation

- (11) Let $f(x) = \sec x$. Find $f''(x)$, the second derivative of $f(x)$.
- A. $2 \sec^2 x \tan x$ B. $\sec x (1 + 2 \tan^2 x)$ C. $2 \tan x (1 + 2 \sec^2 x)$
- D. $2 \sec x^2 \tan^2 x$
- (12) Let $f(x) = \cos^{-1} x$. Find $f'(x)$.
- A. $\frac{1}{1+x^2}$ B. $\frac{1}{\sqrt{1-x^2}}$ C. $\sin^{-1} x$ D. $\frac{\sin x}{1+x^2}$
- (13) Find $g'(3)$ if $g(x) = x^{2^{h(x)}}$ where $h(3) = -2$ and $h'(3) = 5$.
- A. $\frac{1-15 \ln 2}{4}$ B. $\frac{1+15 \ln 2}{4}$ C. $\frac{1-15 \ln 2}{8}$ D. $\frac{1+15 \ln 2}{8}$
- (14) Suppose $x^3 + y^3 - 3xy = 0$. What is $\frac{dy}{dx}$?
- A. $\frac{x^2-y}{x-y}$ B. $\frac{x^2-y}{x-y^2}$ C. $3x^2 + 3y^2 - 3x$ D. None
- (15) If $f(x) = \ln\left(\frac{x^2}{e^x}\right)$, then find $f'(x)$.
- A. $\frac{2}{x} - 1$
- B. $\left(\frac{x^2}{e^x}\right)\left(\frac{2x^2 e^x - x e^x}{e^x}\right)$

- C. $\left(\frac{x^2}{e^x}\right)\left(\frac{2xe^x - 2x^2e^x}{e^x}\right)$
 D. $\frac{2}{x} - 1 + \left(\frac{x^2}{e^x}\right)\left(\frac{2xe^x - xe^x}{e^x}\right)$

(16) If $2y = 1 + xy^3$, then what is the value of $\frac{dy}{dx}$ at the point $(x, y) = (1, 1)$?

- A. 0 B. 1 C. -1 D. ∞

(17) What is y'' if $x^2 + y^2 = 1$?

- A. $2x + 2y$
 B. $-\frac{1}{y^3}$
 C. $2x$
 D. $\frac{2x}{x^2 + y^2}$

(18) What is the McLaurin series expansion for $f(x) = (1 + x)^{\frac{1}{3}}$

- A. $1 + \frac{1}{3}x - \frac{1}{9}x^2 + \dots$
 B. $1 - \frac{1}{3}x + \frac{1}{9}x^2 + \dots$
 C. $1 - \frac{1}{3}x - \frac{1}{9}x^2 + \dots$
 D. $1 + \frac{1}{3}x + \frac{1}{9}x^2 + \dots$

(19) What is the McLaurin series expansion for $f(x) = \sqrt{1 + x}$

- E. $1 + \frac{1}{2}x - \frac{1}{8}x^2 + \dots$
 F. $1 - \frac{1}{2}x + \frac{1}{8}x^2 + \dots$
 G. $1 - \frac{1}{2}x - \frac{1}{8}x^2 + \dots$
 H. $1 + \frac{1}{2}x + \frac{1}{8}x^2 + \dots$

(20) The function y satisfies the equation $4x + 2y = xy^3$. Find the values of $\frac{dy}{dx}$ and of $\frac{d^2y}{dx^2}$ at the point $(x, y) = (1, 2)$.

- A. $y' = -\frac{2}{5}$ and $y'' = \frac{768}{1000}$
 B. $y' = -\frac{2}{5}$ and $y'' = \frac{546}{1000}$
 C. $y' = \frac{2}{5}$ and $y'' = \frac{376}{1000}$
 D. $y' = \frac{2}{5}$ and $y'' = \frac{191}{1000}$

Wherever used below, K is a constant of integration

(21) $\int \frac{1}{1+\sin^2 x} dx$

- A. $\frac{1}{\sqrt{2}} \tan^{-1}(\sqrt{2} \tan x) + K$
- B. $\frac{1}{\sqrt{2}} \sec^{-1}(\sqrt{2} \sec x) + K$
- C. $\frac{1}{\sqrt{2}} \cos^{-1}(\sqrt{2} \cos x) + K$
- D. $\frac{1}{\sqrt{2}} \sin^{-1}(\sqrt{2} \sin x) + K$

(22) $\int_0^1 x^2 \sqrt{1-x^2} dx$

- A. e
- B. 1
- C. π
- D. $\frac{\pi}{16}$

(23) $\int_0^{\pi/2} \frac{\cos x}{2-\cos^2 x} dx$ (Hint: Make substitution $u = \sin x$)

- A. π
- B. $\frac{\pi}{2}$
- C. $\frac{\pi}{3}$
- D. $\frac{\pi}{4}$

(24) $\int_1^2 \frac{\ln x}{(x-3)^3} dx$ (Hint: Integration by parts)

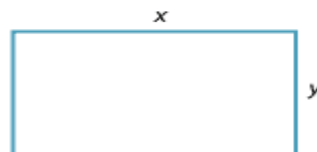
- A. 0
- B. 1
- C. $\ln 2$
- D. $\ln 3$

(25) $\int \frac{x^3}{x^2+2x+5} dx$

- A. $\frac{x^2}{2} - 2x - \ln \sqrt{x^2 + 2x + 5} - \frac{11}{2} \tan^{-1} \frac{1+x}{2} + K$
- B. $\frac{x^2}{2} + 2x - \ln \sqrt{x^2 + 2x + 5} - \frac{11}{2} \tan^{-1} \frac{1+x}{2} + K$
- C. $\frac{x^2}{2} - 2x - \ln \sqrt{x^2 + 2x + 5} + \frac{11}{2} \tan^{-1} \frac{1+x}{2} + K$
- D. $\frac{x^2}{2} - 2x - \ln \sqrt{x^2 + 2x + 5} - \frac{11}{2} \tan^{-1} \frac{1+x}{2} + K$

Part III—Applications of Differentiation and Integration

- (26) Find the dimensions x, y of the rectangle with perimeter $1000m$ so that the area of the rectangle is maximum. (See picture below)



- A. $x = y = 250m$
- B. $x = y = 500m$
- C. $x = y = 200m$
- D. None

Use the function $y = x^2 - 2x + 3$ to answer the next **TWO** questions.

(27) What are the stationary points of y ?

- A. Only $x = 1$
- B. $x = 1$ and $x = 2$
- C. Only $x = 2$
- D. $x = 2$ and $x = -1$

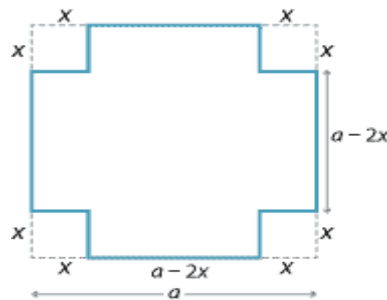
(28) The stationary points of y is/are

- A. Local maxima
- B. Local minima
- C. One is a maximum and the other a minimum
- D. Saddle points

(29) Given the function $y = \frac{x^2}{2} - \cos x$, which of the following statements is true?

- A. $(\frac{\pi^2}{2}, \frac{\pi^2}{8})$ is a local minimum
- B. $(\frac{\pi^2}{2}, \frac{\pi^2}{8})$ is a local maximum
- C. $(0, -1)$ is a local minimum
- D. $(0, -1)$ is a local maximum.

(30) A square sheet of cardboard with each side a centimetres is to be used to make an open-top box by cutting a small square of cardboard from each of the corners and bending up the sides. What is the side length x of the small squares if the box is to have as large a volume as possible? (see picture below)



- A. $x = a$
- B. $x = \frac{a}{6}$
- C. $x = \frac{a}{3}$
- D. $x = \frac{a}{2}$

All the best!

J. Ezeam